

Supplementary Material: Detector Ranking is Not Architecture-Invariant under Medical Covariate Shift

Anonymous CVPR submission

1. Leave-one-backbone W is not a CI

Official Camelyon $W=0.694$ ($n=8$). Dropping one backbone and recomputing W on the remaining seven yields 0.676–0.738 (largest increase when dropping ConvNeXt; largest decrease when dropping EffV2-S or RegNet). This is jackknife *structural* sensitivity of the concordance statistic. It is not a seed confidence interval and is not written as “ $W=0.694$, 95% CI $[\cdot]$.”

2. HED stain jitter

Train-only HED stain jitter ($\sigma=0.20$, $p=1.0$) on ResNet-18, ResNet-50, DenseNet-121, same Adam recipe as official, seed 42. Precommit dirty-guard: DenseNet hospital-2 MSP $<0.80 \Rightarrow$ dirty.

Hospital-2 MSP: ResNet-18 0.772 (enters the 0.695 band), ResNet-50 0.593, DenseNet 0.780 (**below** 0.80). Gap DenseNet–ResNet-18 $=0.008$. id-val acc ≥ 0.994 on all three. Coverage@risk 10% saturates at 1.0 for these three (not worse than official ResNet-18 0.145). Held-out hospital-1 MSP: ResNet-18 0.765, ResNet-50 0.685, DenseNet 0.796.

Dirty. Part of the apparent gap shrink is DenseNet falling (0.883 $\rightarrow$ 0.780) while ResNet-18 rises. Same class as CE+SupCon. Not an open mitigation. Not expanded to eight CNNs.

3. MIDOG in-house zoo $n=8$ (not official W)

Official MIDOG W remains $n=3$, 0.857. Five additional in-house CNNs plus the original three, same 1a vs. 1b+1c protocol, *not* mixed with public OpenMIBOOD R50 ~ 0.59 :

Backbone	MSP	Maha
ResNet-18	0.438	0.733
ResNet-50	0.512	0.839
DenseNet-121	0.447	0.910
ConvNeXt-Tiny	0.521	0.801
MobileNetV3-L	0.469	0.714
RegNetY-3.2GF	0.553	0.944
EfficientNet-B3	0.486	0.673
EfficientNetV2-S@224	0.493	0.581

MSP 0.44–0.55 on all eight; 0/6 non-ResNet jumps versus that domain’s ResNet mean. Consistent with official $n=3$: no Camelyon-style MSP jump. A domain-scope bound, not a replacement of official W .

4. Shift-type extract

Same eight Camelyon seed-42 classifiers; ID = Camelyon id_val. Arm C = hospital-2 (frozen). Arm T = MIDOG 1a (task-shift H&E). Arm F = CIFAR-10 test (far-OOD). Extract only; T/F cells are not mixed into official W .

Arm	non-ResNet jump	readout
C covariate	6/6	seed-42 count, not multi-seed
T task-shift	0/6	jump disappears
F far-OOD	2/6	mixed

Joint: **no law.** Per-arm Kendall W (8 backbones $\times$ 7 methods, never pooled, not official W): C 0.731, T 0.692, F 0.846.

5. Five-seed jumps

Jump = MSP minus that seed’s ResNet mean. Threshold 0.15. Official W is seed 42 only. The MSP matrix is Table 2 of the main paper.

seed	R mean	Dense	ConvNeXt	Mobile	RegNet	EffB3	EffV2
42	0.545	+0.338	+0.301	+0.226	+0.239	+0.258	+0.234
43	0.667	+0.199	+0.133	+0.061	+0.112	+0.064	+0.077
44	0.610	+0.258	+0.193	+0.205	+0.226	+0.033	+0.198
45	0.520	+0.325	+0.244	+0.260	+0.310	+0.298	+0.327
46	0.621	+0.205	+0.164	+0.124	+0.103	+0.057	+0.214

6. Other closed checks

ECE, BN-drift, eigenspace summaries, n_{train} sweeps, and CIFAR-10-C were closed earlier and are not pillars. CIFAR semantic/far-OOD is not a main-domain result.

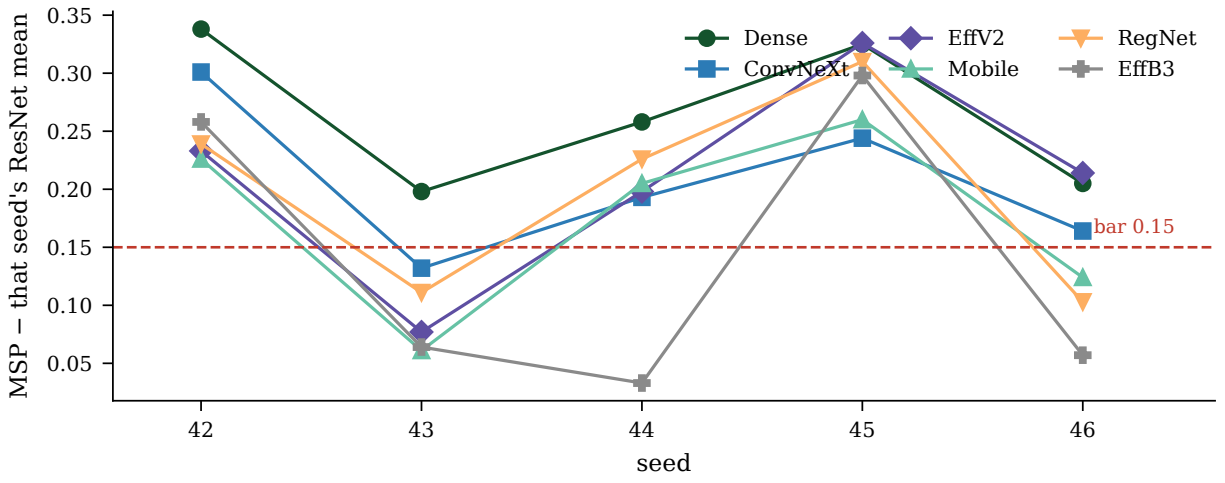


Figure 1. Per-seed MSP jump versus that seed’s ResNet mean. DenseNet stays above 0.15 on all five seeds; EfficientNet-B3 does not.